

API — Airfare Price Index for India (0.1.0)

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Real-Time Airfare Price Index for India (API) — SIH 2026 PS 26056 (MoSPI).

Every published value carries `n_obs` and `coverage_pct`, and every response carries the `pipeline_run` provenance stamp (code version, git SHA, methodology config hash, input row count) needed to reproduce it.

Methodology proof of concept. Not an official government statistic.

index

Headline

QUERY PARAMETERS

dow_adjusted	boolean (Dow Adjusted) Default: <code>false</code> Return the day-of-week-adjusted series.
freq	string (Freq) Default: <code>"daily"</code> daily weekly monthly
from >	From (string) or From (null) (From) Inclusive start date.
to >	To (string) or To (null) (To) Inclusive end date.

Responses

✓ 200 Successful Response

RESPONSE SCHEMA: application/json

series required	string (Series)
--------------------	-----------------

⌵ dow_adjusted required	boolean (Dow Adjusted)
⌵ base_period > required	object (BasePeriod)
⌵ pipeline_run > required	object (PipelineRunModel)
⌵ data_mode > required	object (DataModeSummary) Real-vs-synthetic lineage. Present on every index response by design. A consumer must never have to guess whether a series is a measurement or a simulation, so this travels with the data rather than living in a README.
⌵ count required	integer (Count)
⌵ points > required	Array of objects (Points)
⌵ freq	string (Freq) Default: <code>"daily"</code> daily weekly monthly

⌵ 422 Validation Error

RESPONSE SCHEMA: application/json

⌵ detail > Array of objects (Detail)

GET /api/v1/index

Response samples

200

422

Content type

application/json

Copy Expand all Collapse all

```
{
  - "base_period": {
    "end": "string",
    "n_days": 0,
    "start": "string"
  },
  "count": 0,
```

```

- "data_mode": {
  "banner": "string",
+ "counts": { ... },
  "is_demo_data": true,
  "real_share": 0,
  "synthetic_share": 0,
  "total_rows": 0
},
"dow_adjusted": true,
"freq": "daily",
- "pipeline_run": {
  "code_version": "string",
  "config_hash": "string",
  "created_at": "string",
  "git_sha": "string",
  "index_eligible_rows": 0,
  "input_row_count": 0,
  "run_id": "string"
},
- "points": [
  + { ... }
],
"series": "string"
}

```

Leadtime

Index by advance-purchase window (base period = 100): how fast each window's fares are inflating. This is NOT the fare-level curve — see </leadtime/curve>.

Responses

✓ **200** Successful Response

RESPONSE SCHEMA: application/json

note required	string (Note)
pipeline_run > required	object (PipelineRunModel)

└─ windows > required	Array of objects (Windows)
---	----------------------------

GET /api/v1/index/leadtime



Response samples

200

Content type

application/json

Copy Expand all Collapse all

```
{
  "note": "string",
  - "pipeline_run": {
    "code_version": "string",
    "config_hash": "string",
    "created_at": "string",
    "git_sha": "string",
    "index_eligible_rows": 0,
    "input_row_count": 0,
    "run_id": "string"
  },
  - "windows": [
    + { ... }
  ]
}
```

Leadtime Curve

Relative fare LEVEL by advance window (14-day window = 100): how much more a last-minute booking costs than an advance one.

QUERY PARAMETERS

└─ as_of >	As Of (string) or As Of (null) (As Of) Curve as at this date; latest if omitted.
-------------------------------	---

Responses

✓ 200 Successful Response

RESPONSE SCHEMA: application/json

as_of > required	As Of (string) or As Of (null) (As Of)
reference_window > required	Reference Window (integer) or Reference Window (null) (Reference Window)
curve > required	Array of objects (Curve)
note >	Note (string) or Note (null) (Note)

✓ 422 Validation Error

RESPONSE SCHEMA: application/json

detail >	Array of objects (Detail)
----------	---------------------------

GET

/api/v1/index/leadtime/curve

▼

Response samples

200

422

Content type

application/json

Copy

Expand all

Collapse all

```
{
  "as_of": "string",
  - "curve": [
    + { ... }
  ],
  "note": "string",
  "reference_window": 0
}
```

Routes

Responses

✓ 200 Successful Response

RESPONSE SCHEMA: application/json

pipeline_run > required	object (PipelineRunModel)
count required	integer (Count)
routes > required	Array of objects (Routes)

GET /api/v1/index/routes

Response samples

200

Content type
application/json

Copy Expand all Collapse all

```
{
  "count": 0,
  - "pipeline_run": {
    "code_version": "string",
    "config_hash": "string",
    "created_at": "string",
    "git_sha": "string",
    "index_eligible_rows": 0,
    "input_row_count": 0,
    "run_id": "string"
  },
  - "routes": [
    + { ... }
  ]
}
```

Route Heatmap

Route x date matrix for the sector-wise heatmap.

Declared BEFORE `/index/routes/{route_code}` so the literal path wins the route match — otherwise 'heatmap' is captured as a route code and returns 404.

QUERY PARAMETERS

from >	From (string) or From (null) (From)
to >	To (string) or To (null) (To)

Responses

✓ 200 Successful Response

RESPONSE SCHEMA: application/json

routes required	Array of strings (Routes)
route_names required	Array of strings (Route Names)
dates required	Array of strings (Dates)

matrix > required	Array of numbers or null (Matrix) [items] matrix[i][j] = routes[i] on dates[j]; null = no index, NOT zero.
value_min > required	Value Min (number) or Value Min (null) (Value Min)
value_max > required	Value Max (number) or Value Max (null) (Value Max)
baseline required	number (Baseline)
note required	string (Note)
data_mode > required	object (DataModeSummary) Real-vs-synthetic lineage. Present on every index response by design. A consumer must never have to guess whether a series is a measurement or a simulation, so this travels with the data rather than living in a README.

▼ 422 Validation Error

RESPONSE SCHEMA: application/json

→ detail > Array of objects (Detail)

GET /api/v1/index/routes/heatmap

Response samples

200

422

Content type

application/json

Copy Expand all Collapse all

```
{
  "baseline": 0,
  - "data_mode": {
    "banner": "string",
    + "counts": { ... },
    "is_demo_data": true,
    "real_share": 0,
    "synthetic_share": 0,
    "total_rows": 0
  },
}
```

```
- "dates": [  
    "string"  
  ],  
- "matrix": [  
    + [ ... ]  
  ],  
  "note": "string",  
- "route_names": [  
    "string"  
  ],  
- "routes": [  
    "string"  
  ],  
  "value_max": 0,  
  "value_min": 0  
}
```

Route

PATH PARAMETERS

 route_code required	string (Route Code)
---	---------------------

Responses

✓ **200 Successful Response**

RESPONSE SCHEMA: application/json

 route_code required	string (Route Code)
 display_name required	string (Display Name)
 weight required	number (Weight)
 pipeline_run > required	object (PipelineRunModel)
 count required	integer (Count)
 points > required	Array of objects (Points)

RESPONSE SCHEMA: application/json

→ detail >

Array of objects (Detail)

GET

/api/v1/index/routes/{route_code}



Response samples

200

422

Content type

application/json

Copy

Expand all

Collapse all

```
{
  "count": 0,
  "display_name": "string",
  - "pipeline_run": {
    "code_version": "string",
    "config_hash": "string",
    "created_at": "string",
    "git_sha": "string",
    "index_eligible_rows": 0,
    "input_row_count": 0,
    "run_id": "string"
  },
  - "points": [
    + { ... }
  ],
  "route_code": "string",
  "weight": 0
}
```

Volatility

Daily and intraday fare volatility, plus the sparse-sampling measurement-error analysis: the evidence that daily collection is worth its cost.

Responses

✓ 200 Successful Response

RESPONSE SCHEMA: application/json

→ property name*
additional property any

GET /api/v1/index/volatility

Response samples

200

Content type
application/json

{ }

Copy

methodology

Methodology

The basket, the index-number formulae, the methodology fingerprint, and the full cleaning row-accounting. Publishing this is what separates a statistic from a number.

Responses

✓ 200 Successful Response

RESPONSE SCHEMA: application/json

→ property name*
additional property

any

GET

/api/v1/methodology

▼

Response samples

200

Content type

application/json

Copy

{ }

Pipeline Run

Responses

✓ 200 Successful Response

RESPONSE SCHEMA:

application/json

→ run_id required	string (Run Id)
→ code_version required	string (Code Version)
→ git_sha required	string (Git Sha)
→ config_hash required	string (Config Hash)
→ input_row_count required	integer (Input Row Count)
→ index_eligible_rows required	integer (Index Eligible Rows)

created_at required	string (Created At)
---	---------------------

GET /api/v1/pipeline-run



Response samples

200

Content type

application/json

Copy

```
{
  "code_version": "string",
  "config_hash": "string",
  "created_at": "string",
  "git_sha": "string",
  "index_eligible_rows": 0,
  "input_row_count": 0,
  "run_id": "string"
}
```

reference

Route Metadata

Route dimension for dropdowns and filters. Identity only, no series.

Responses

✓ 200 Successful Response

RESPONSE SCHEMA: application/json

```
count
required

routes >
required
```

integer (Count)

Array of objects (Routes)

GET /api/v1/routes



Response samples

200

Content type

application/json

Copy Expand all Collapse all

```
{
  "count": 0,
  "routes": [
    + { ... }
  ]
}
```

validation

Validation Dgca

Back-test against the reference series, with lineage stated up front.

Always returns 200 with an `available` flag rather than erroring when no reference is loaded: "no reference yet" is an ordinary operational state early in collection, not a client mistake, and the dashboard needs to render something either way.

Responses

✓ 200 Successful Response

RESPONSE SCHEMA: application/json

property name*
additional property any

GET /api/v1/validation/dgca

Response samples

200

Content type
application/json

Copy

{ }

ops

Health

Responses

200 Successful Response

RESPONSE SCHEMA: application/json

status required	string (Status)
data_available required	boolean (Data Available)
latest_index_date > required	Latest Index Date (string) or Latest Index Date (null) (Latest Index Date)

code_version required	string (Code Version)
data_mode >	DataModeSummary (object) or null
hours_since_latest_index >	Hours Since Latest Index (number) or Hours Since Latest Index (null) (Hours Since Latest Index) Age of the most recent index point. A stale series looks identical to a flat one on a chart, so the age is published rather than inferred.

GET /health

Response samples

200

Content type
application/json

Copy Expand all Collapse all

```
{
  "code_version": "string",
  "data_available": true,
  - "data_mode": {
    "banner": "string",
    + "counts": { ... },
    "is_demo_data": true,
    "real_share": 0,
    "synthetic_share": 0,
    "total_rows": 0
  },
  "hours_since_latest_index": 0,
  "latest_index_date": "string",
  "status": "string"
}
```